

Random Forest Regression: Numerical Worked Example

1 Problem Statement

Given the following dataset with one input feature x and output y , use **Random Forest Regression** with **3 decision trees (stumps)** to predict the value of y for $x = 3.5$.

2 Dataset

Sample	x	y
1	1	2
2	2	3
3	3	5
4	4	4
5	5	6

Table 1: Original Dataset

3 Tree 1

Bootstrap Sample

x	y
1	2
2	3
3	5
5	6
5	6

Split

$$x \leq 3 \quad \text{vs} \quad x > 3$$

Leaf Predictions

Left leaf ($x \leq 3$):

$$\hat{y}_L = \frac{2 + 3 + 5}{3} = 3.33$$

Right leaf ($x > 3$):

$$\hat{y}_R = \frac{6 + 6}{2} = 6$$

Prediction for $x = 3.5$

$$x = 3.5 > 3 \Rightarrow \hat{y}_1 = 6$$

4 Tree 2

Bootstrap Sample

x	y
2	3
3	5
4	4
4	4
5	6

Split

$$x \leq 4 \quad \text{vs} \quad x > 4$$

Leaf Predictions

Left leaf ($x \leq 4$):

$$\hat{y}_L = \frac{3 + 5 + 4 + 4}{4} = 4$$

Right leaf ($x > 4$):

$$\hat{y}_R = 6$$

Prediction for $x = 3.5$

$$x = 3.5 \leq 4 \Rightarrow \hat{y}_2 = 4$$

5 Tree 3

Bootstrap Sample

x	y
1	2
3	5
3	5
4	4
5	6

Split

$$x \leq 2.5 \quad \text{vs} \quad x > 2.5$$

Leaf Predictions

Left leaf ($x \leq 2.5$):

$$\hat{y}_L = 2$$

Right leaf ($x > 2.5$):

$$\hat{y}_R = \frac{5 + 5 + 4 + 6}{4} = 5$$

Prediction for $x = 3.5$

$$x = 3.5 > 2.5 \Rightarrow \hat{y}_3 = 5$$

6 Random Forest Prediction

Random Forest regression predicts the output by averaging predictions from all trees.

Tree	Prediction
Tree 1	6
Tree 2	4
Tree 3	5

$$\hat{y} = \frac{6 + 4 + 5}{3} = \boxed{5}$$

7 Conclusion

- Each tree is trained on a bootstrap sample.
- Decision stumps predict the mean value of the target.
- Random Forest regression averages predictions from all trees.
- Final predicted value for $x = 3.5$ is $\boxed{5}$.